

Disproof of conjecture 00000002718: the boundary of the tetrahedron is shellable but not 2-collapsible

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1 The conjecture and the clause refuted

Statement (English, verbatim). Definition: d -collapsible complexes: reduced by matched contractions of d -faces. Conjecture: The separation degree of d -collapsibility from shellability: shellable implies d -collapsible but not conversely; the minimal separating complex is a special 2-complex, and the separation spectrum is a finite layer in dimension two. (d -collapsible separation spectrum)

The Chinese text gives the same definition (reduction by matched contractions of d -faces) and the same claims.

Reading. The conjecture is a conjunction. Its first conjunct is

(S) every shellable d -dimensional (finite) simplicial complex is d -collapsible.

Here d is the dimension of the complex, as the phrase “ d -faces” indicates. We refute (S) for $d = 2$, even for pure complexes, which refutes the conjunction. We do not address the other conjuncts (“not conversely”, the “minimal separating complex” and the “separation spectrum”).

Convention on d . Every finite complex of dimension less than d is d -collapsible, since each maximal face τ has $\dim \tau \leq d - 1$ and can be removed by itself (a remark, not formalized). So an existential reading that lets d be chosen large would be trivial; the universal claim (S), however, is not, and our counterexample needs no global restriction on d : it takes $d = 2 = \dim T$, and we also show the witness is not d -collapsible for $d = 0, 1$.

2 Definitions

All complexes are finite abstract simplicial complexes: a finite family K of finite sets closed under taking subsets. The dimension of a face σ is $|\sigma| - 1$. A face is *maximal* if no other face contains it. K is *pure of dimension d* if it is nonempty, every face has dimension at most d , and every maximal face has dimension exactly d .

Definition 1 (d -collapsibility, Wegner). The notion was introduced by G. Wegner, *d -Collapsing and nerves of families of convex sets*, Arch. Math. 26 (1975) 317–321 (<https://doi.org/10.1007/bf01229745>). We use the precise form stated by M. Tancer, *d -collapsibility is NP-complete for d greater or equal to 4* (arXiv:0808.1991). Verbatim: “A face σ of a simplicial complex K is collapsible if there is a unique maximal face of K containing σ ... we denote this maximal face by $\tau(\sigma)$. (We allow $\tau(\sigma) = \sigma$.) Moreover, if $\dim \sigma \leq d - 1$, then σ is d -collapsible. By $[\sigma, \tau(\sigma)]$ we denote the set $\{\eta \in K \mid \sigma \subseteq \eta \subseteq \tau(\sigma)\}$... the complex $K' = K \setminus [\sigma, \tau(\sigma)]$ arises from K by an elementary d -collapse.

... a complex K is d -collapsible if $K \rightarrow \emptyset$." Here $\rightarrow$ is a finite sequence of elementary d -collapses. In Lean: `ElemDCollapse`, `DCollapsesTo` (reflexive transitive closure), `DCollapsible`.

Definition 2 (Shelling). From Wikipedia, *Shelling (topology)* ([https://en.wikipedia.org/wiki/Shelling_\(topology\)](https://en.wikipedia.org/wiki/Shelling_(topology))): "An ordering $C_1, C_2, \dots$ of the maximal simplices of Δ is a shelling if, for all $k = 2, 3, \dots$, the complex $B_k := (\bigcup_{i=1}^{k-1} C_i) \cap C_k$ is pure and of dimension one smaller than $\dim C_k$." Here C_i stands for the complex of all subsets of C_i . In Lean, `IsShelling K L` says that the list L has no duplicates, lists exactly the maximal faces of K , and, for every 0-based index $k \geq 1$, the family `shellBoundary L k C_k` = $\{\eta \subseteq C_k \mid \exists i < k, \eta \subseteq C_i\}$ is nonempty and all its maximal faces have $|C_k| - 1$ elements. `Shellable K` means that some shelling exists.

Definition 3 (Collapse-type moves, for robustness). The gloss "matched contractions of d -faces" is not a standard definition. To show the result does not depend on it, we also use Whitehead's collapses as stated in Wikipedia, *Collapse (topology)* ([https://en.wikipedia.org/wiki/Collapse_\(topology\)](https://en.wikipedia.org/wiki/Collapse_(topology))). If $\tau \subsetneq \sigma$, σ is a maximal face and no other maximal face contains τ , then " τ is called a free face. A simplicial collapse of K is the removal of all simplices γ such that $\tau \subseteq \gamma \subseteq \sigma$, where τ is a free face. If additionally we have $\dim \tau = \dim \sigma - 1$, then this is called an elementary collapse. A simplicial complex that has a sequence of collapses leading to a point is called collapsible." In Lean these are `IsFreeFace`, `SimplicialCollapse`, `ElementaryCollapse` and `Collapsible` (collapse to $\{\emptyset, \{v\}\}$). These auxiliary Lean definitions use an augmented convention: they also allow the empty face as a free face (so, e.g., $\{\emptyset, \{v\}\}$ may collapse to the void complex), which ordinary homotopy-preserving Whitehead collapses exclude. This only makes the moves more permissive; the obstruction below holds even under it, since T has no free face of any dimension.

3 The witness and the proof

Let T be the boundary of the tetrahedron: all proper subsets of $\{0, 1, 2, 3\}$ (15 faces, including $\emptyset$). In Lean, `tetraBoundary := (univ : Finset (Fin 4)).powerset.erase univ`. Its maximal faces are the four triangles $\{0, 1, 2\}, \{0, 1, 3\}, \{0, 2, 3\}, \{1, 2, 3\}$, so T is a simplicial complex, pure of dimension 2.

Lemma 4 (Shelling). $C_1 = \{0, 1, 2\}$, $C_2 = \{0, 1, 3\}$, $C_3 = \{0, 2, 3\}$, $C_4 = \{1, 2, 3\}$ is a shelling of T .

Proof. B_2 is generated by the edge $\{0, 1\}$; B_3 by the edges $\{0, 2\}, \{0, 3\}$; B_4 by the edges $\{1, 2\}, \{1, 3\}, \{2, 3\}$. Each is pure of dimension $1 = \dim C_k - 1$. (Lean: `tetraBoundary_shelling`, checked by `decide` on the finite encoding.) $\square$

Lemma 5 (No free face). *Every face of T that is not maximal lies in at least two maximal faces. So T has no free face.*

Proof. An edge lies in 2 triangles, a vertex in 3, and $\emptyset$ in all 4 (Lean: `two_facets_above`, by `decide`). A non-maximal face σ properly contained in a triangle has $|\sigma| \leq 2$, so it lies in two distinct triangles, and no triangle is the unique maximal face containing it (Lean: `tetraBoundary_no_free_face`). $\square$

Lemma 6 (Stuck complexes). *Let R be any move relation on complexes such that every R -move out of K uses a free face of K . If K has no free face, then every complex reachable from K by finitely many R -moves equals K . (Lean: `stuck_of_no_free_face`.)*

Simplicial and elementary collapses use a free face by definition (`simplicialCollapse_needs_free`, `elementaryCollapse_needs_free`). An elementary d -collapse uses a face σ with $|\sigma| \leq d$ and a unique maximal face $\tau(\sigma)$. If every maximal face has more than d vertices, then $\sigma \neq \tau(\sigma)$, so σ is free (`elemDCollapse_needs_free`). In T every maximal face has 3 vertices, so this applies for every $d \leq 2$.

Theorem 7. *For every $d \leq 2$, T is not d -collapsible. Also, T admits no simplicial or elementary collapse, and T is not collapsible.*

Proof. By the two lemmas, any d -collapsing of T ($d \leq 2$) ends at T itself (`dCollapsesTo_eq`). But $T \neq \emptyset$, so $T \not\approx \emptyset$ (`tetraBoundary_not_dCollapsible`). In the same way, every sequence of simplicial or elementary collapses ends at T (`tetraBoundary_simplicialCollapse_stuck`, `tetraBoundary_elementaryCollapse_stuck`). T has 15 faces and a point has 2, so T is not collapsible (`tetraBoundary_not_collapsible`). $\square$

Theorem 8 (Main). *Statement (S) is false for $d = 2$. In Lean:*

$$\neg \forall (n \ d : \mathbb{N}) (K : \text{Finset } (\text{Finset } (\text{Fin } n))), \quad \text{IsComplex } K \rightarrow \text{IsPureOfDim } K \ d \\ \rightarrow \text{Shellable } K \rightarrow \text{DCollapsible } d \ K$$

(theorem `C2718.conjecture2718_false`) together with counterexample: `IsComplex tetraBoundary`, `IsPureOfDim tetraBoundary 2`, `Shellable tetraBoundary` and `¬DCollapsible 2 tetraBoundary`.

4 Lean formalization and axiom audit

The file `lean/Conjecture2718/Basic.lean` (237 lines, import `Mathlib` only, Lean 4.33.1, `Mathlib v4.33.1`) contains all definitions above for an arbitrary vertex type with decidable equality. The finite facts about T are checked by the kernel with `decide` on the faithful encoding `Finset (Finset (Fin 4))`. The output of `#print axioms` for `C2718.conjecture2718_false`, `C2718.counterexample`, `C2718.tetraBoundary_not_collapsible` and `C2718.tetraBoundary_elementaryCollapse_stuck` is `[propext, Classical.choice, Quot.sound]` in each case. There is no `sorry`, no `native_decide` and no added axiom.